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Display unit and vehicular display device

Abstract

There are a display unit and a vehicular display device. The display unit emits a display image including display light to be projected onto a projection target member provided in a vehicle. The display unit includes: a lens including a lens body and a first support portion protruding from an edge portion of the lens body; and a case that has a space configured to accommodate the first support portion and holds the lens by fitting the first support portion into the space.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is based on and claims priority under 35 USC 119 from Japanese Patent Application No. 2023-079591 filed on May 12, 2023, the contents of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to a display unit and a vehicular display device.

BACKGROUND ART

(3) In recent years, a vehicle such as an automated vehicle is equipped with a vehicular display device such as a head-up display (HUD). In the vehicular display device, for example, a display image displayed on a display unit is projected onto a windshield, a combiner, or the like via an optical system such as a reflection mirror, so that a driver visually recognizes the display image as a virtual image. See JP2022-160754A, for example.

(4) In a vehicular display device in the related art, when a lens is assembled, positioning holes are respectively provided in two support portions of the lens, and the lens is assembled in a manner of fitting positioning pins provided in a case into the positioning holes. However, it is required to ensure an area corresponding to sizes of the positioning holes in the support portions of the lens, and as the area is ensured, the lens is increased in size.

SUMMARY OF INVENTION

(5) The present disclosure provides a display unit and a vehicular display device that can be reduced in size.

(6) According to an illustrative aspect of the present disclosure, a display unit emits a display image including display light to be projected onto a projection target member provided in a vehicle. the display unit includes: a lens including a lens body and a first support portion protruding from an edge portion of the lens body; and a case that has a space configured to accommodate the first support portion and holds the lens by fitting the first support portion into the space.

(7) According to another illustrative aspect of the present disclosure, a vehicular display device includes: the above display unit; and an optical component configured to reflect the display light emitted from the display unit.

(8) According to the present disclosure, the display unit and the vehicular display device can be reduced in size.

(9) The present disclosure is briefly described above. Further, details of the present disclosure will be further clarified by reading modes for carrying out the disclosure described below with reference to the accompanying drawings.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a schematic diagram illustrating a schematic configuration of a vehicle equipped with a vehicular display device.

(2) FIG. 2 is a schematic diagram illustrating a schematic configuration of the vehicular display device.

(3) FIG. 3 is an exploded perspective view illustrating a backlight unit according to an embodiment.

(4) FIG. 4 is an exploded perspective view illustrating lenses.

(5) FIG. 5 is a cross-sectional view illustrating the backlight unit according to the embodiment.

(6) FIG. 6 is an exploded perspective view illustrating a backlight unit in the related art.

(7) FIG. 7 is a cross-sectional view illustrating the backlight unit in the related art.

DESCRIPTION OF EMBODIMENTS

(8) Hereinafter, a display unit and a vehicular display device according to an embodiment of the

present disclosure will be described with reference to the drawings.

(9) As illustrated in FIG. 1, a vehicular display device **1** including a display unit is a head-up display device mounted on a vehicle **100** such as an automated vehicle. The vehicular display device **1** is disposed inside an instrument panel **102** in the vehicle **100**, and projects a display image onto a windshield **103** that is an example of a projection target member. Since the windshield **103** has a semi-transparent property of reflecting a part of incident light and transmitting remaining light, the windshield **103** reflects display light L emitted from the vehicular display device **1** as a display image toward an eye point EP of a driver D while transmitting foreground of the vehicle **100**. The driver D is an occupant of the vehicle **100**, a driver in particular, and recognizes the display image reflected by the windshield **103** as a virtual image S. The virtual image S is recognized by the driver D in front of the windshield **103**. As illustrated in FIG. 2, the vehicular display device **1** according to the present embodiment includes a housing **2**, two reflection mirrors **3**, a backlight unit **5**, and a control unit **7**.

(10) The housing **2** is formed of, for example, synthetic resin, and is fixed to a vehicle body (not illustrated). As illustrated in FIG. 2, the housing **2** accommodates and supports the backlight unit **5**, the two reflection mirrors **3**, and the control unit **7** in an internal space **2b**. The housing **2** has an opening **2a** through which the outside communicates with the internal space **2b**. The opening **2a** is provided in the housing **2** at a position facing the windshield **103**, and is closed by a cover member **6**. The display light L emitted from the backlight unit **5** and reflected by the two reflection mirrors **3** is transmitted through the cover member **6**. The display light L transmitted through the cover member **6** travels toward the windshield **103**.

(11) The two reflection mirrors **3** are disposed on an optical path of the display light L from the backlight unit **5** to the windshield **103**, and reflect the display light L emitted from the backlight unit **5** toward the windshield **103**. The two reflection mirrors **3** include a plane mirror **8** serving as an example of an optical component and a concave mirror **9** serving as an example of an optical component.

(12) The plane mirror **8** has a reflection surface formed as a plane and is disposed at a position facing the backlight unit **5**. The reflection surface of the plane mirror **8** totally reflects the display light L emitted from the backlight unit **5** toward the concave mirror **9**.

(13) The concave mirror **9** has a reflection surface formed as a concave curved surface and is disposed at a position facing the plane mirror **8**. The concave mirror **9** totally reflects the display light L reflected by the plane mirror **8** toward the windshield **103** via the cover member **6**. For example, the concave mirror **9** has a function of a magnifying mirror. The concave mirror **9** enlarges and reflects a display image represented by the display light L reflected by the concave mirror **9** such that the display image is relatively larger than a display image represented by the display light L before being reflected by the concave mirror **9**.

(14) The backlight unit **5** is an example of a display unit, and emits, as the display light L, a display image to be visually recognized as the virtual image S by the driver D of the vehicle **100**. The backlight unit **5** includes a display panel **55** (see FIG. 3) and a light source (not illustrated) in a housing having an opening. The backlight unit **5** may include a component such as a condenser lens and a deflector lens. The backlight unit **5** is a display unit that emits, as display light, a display image to be projected onto the windshield **103** that is a projection target member.

(15) The control unit **7** is connected to the backlight unit **5** and controls the backlight unit **5**. The control unit **7** includes, for example, an IC chip mounted on a substrate, and is driven by electric power obtained from a battery mounted on the vehicle **100**.

(16) As illustrated in FIG. 3, the backlight unit **5** includes lenses **40** and a case **70** that accommodates the lenses **40**. The lens **40** is formed of a material such as glass and transparent resin, and plays a role of refracting and deflecting light emitted from a light source (not illustrated) toward the display panel **55** provided on an upper case **50** to be described later.

(17) The lens **40** includes a lens body **41**, a first support portion **43**, and a second support portion

45. The lens body **41** has a main lens function, has a rectangular shape when viewed in an optical axis direction, and deflects light incident from a light source (not illustrated). The first support portion **43** is a plate-shaped member that is an edge portion of the lens body **41** and protrudes from one side. The second support portion **45** is a tab-shaped member provided on the other side opposite to the one side where the first support portion **43** is provided, and has a positioning hole **47**.

(18) The case **70** includes the upper case **50** serving as a first case and a lower case **60** serving as a second case. The case **70** is completed by combining the upper case **50** and the lower case **60**, and the lenses **40** are accommodated in the case **70**.

(19) The upper case **50** has four side walls and is implemented in a lid shape. A first side wall **51** which is one of the four side walls has an opening **53** formed by cutting a part of the first side wall **51**. The opening **53** defines a support portion fitting space R to be described later.

(20) The lower case **60** also has four side walls, and is formed in a box shape. A second side wall **62** which is one of the four side walls has a placement surface **64** extending outward from an upper end portion of the second side wall **62**. A pair of positioning pins **66** protruding from the placement surface **64** toward the upper case **50** are provided at both ends of the placement surface **64**. Two ribs **68** are provided between the pair of positioning pins **66** on the placement surface **64**. Further, a positioning pin **69** is provided on a side wall facing the second side wall **62**.

(21) The lenses **40** are disposed inside the lower case **60** when assembling the backlight unit **5**. At this time, the first support portion **43** of the lens **40** is placed on the placement surface **64** of the lower case **60**, and the pair of positioning pins **66** function as a pair of position regulating portions for regulating a position of the first support portion **43** on the placement surface **64**.

(22) Further, the positioning pin **69** of the lower case **60** is inserted into the positioning hole **47** formed in the second support portion **45** of the lens **40**. The positioning pin **69** functions as a positioning portion that positions the lens **40**.

(23) Thereafter, the upper case **50** covers the lower case **60** and the lenses **40**. The opening **53** is provided in the first side wall **51** of the upper case **50**. During assembly, the opening **53** is formed at a position facing the placement surface **64** of the lower case **60**. Therefore, the case **70** completed by combining the upper case **50** and the lower case **60** has a space capable of accommodating the first support portion **43** of the lens **40**, that is, the support portion fitting space R. The case **70** can hold the lens **40** by fitting the first support portion **43** into the support portion fitting space R.

(24) As described above, according to the backlight unit **5** in the present embodiment, the lens **40** is positioned and held in the case **70** by fitting the first support portion **43** into the support portion fitting space R of the case **70**. Accordingly, as compared with a case where positioning holes are provided in support portions of the lens **40** and the lens **40** is assembled in a manner of fitting positioning pins provided in the case **70** into the positioning holes, an area of a support portion can be reduced by a size of a positioning hole. Accordingly, since it is not necessary to ensure this area, the lens **40** can be reduced in size and the backlight unit **5** can be reduced in size.

(25) Since no positioning hole is provided in the support portion (the first support portion **43**) of the lens **40**, formability of the lens **40** is improved, and the degree of freedom in designing a shape of the lens is increased. This makes it easy to adopt an optimum shape for improving visibility. Further, it is possible to achieve a Fresnel lens which cannot be achieved in a lens in the related art in which positioning holes are provided in support portions. Therefore, the lens **40** can be further reduced in size and thickness by adopting the Fresnel lens.

(26) In the present embodiment, the case **70** includes the upper case **50** and the lower case **60**, and the support portion fitting space R is defined by combining the upper case **50** and the lower case **60**. Accordingly, the first support portion **43** of the lens **40** is positioned between the upper case **50** and the lower case **60**, and the case **70** sandwiches the first support portion **43** in the support portion fitting space R by combining the upper case **50** and the lower case **60**. Accordingly, the lens

40 can be easily assembled to the case **70**.

(27) In particular, the support portion fitting space **R** is defined by the opening portion **53**, the placement surface **64**, and the pair of positioning pins **66** in a state where the upper case **50** and the lower case **60** are combined. With such a configuration, since a position of the first support portion **43** placed on the placement surface **64** is regulated by the pair of positioning pins **66**, assembling accuracy of the lens **40** to the case **70** can be improved. In particular, the pair of positioning pins **66** can regulate a position of a side end portion of the first support portion **43** and regulate a lateral movement of the first support portion **43**.

(28) The lower case **60** is provided with the positioning pin **69** to be inserted into the positioning hole **47** of the lens **40**. Accordingly, the upper case **50** and the lower case **60** are combined in a state where the first support portion **43** on one side of the rectangular lens body **41** is fitted into the support portion fitting space **R** and the positioning pin **69** is inserted into the positioning hole **47** in the second support portion **45** on the other side. Therefore, the lens **40** can be reduced in size as compared with a case where positioning holes are provided in support portions on one side and the other side of the lens body **41**. The positioning pin **69** serving as a positioning portion may be provided not only in the lower case **60**, but also in either the upper case **50** or the lower case **60**.

(29) Further, the two ribs **68** provided on the placement surface **64** abut against a lower face of the first support portion **43** and stably hold the first support portion **43**.

(30) As illustrated in FIGS. **4** and **5**, two lenses **40** are used in the backlight unit **5** in the present embodiment. The two lenses **40** are laminated, the first support portion **43** of the upper lens **40** and the first support portion **43** of the lower lens **40** are integrated, and the second support portion **45** of the upper lens **40** and the second support portion **45** of the lower lens **40** are integrated. When two or more lenses **40** are used, for example, a magnification ratio can be increased, and thus improvement in visibility can be expected.

(31) When a plurality of lenses **40** are used to improve visibility, the backlight unit **5** may be increased in size. According to such a configuration, since the lens **40** can be reduced in size, a plurality of lenses **40** can be used without increasing a size of a case of the backlight unit **5**, and display quality of the vehicular display device **1** can be improved. Of course, only one lens **40** may be used.

(32) As illustrated in FIG. **2**, the vehicular display device **1** includes at least the backlight unit **5** according to the present embodiment and optical components that reflect display light emitted from the backlight unit **5**, that is, the plane mirror **8** and the concave mirror **9**. Accordingly, the vehicular display device **1** can be reduced in size and display quality can be improved.

(33) As illustrated in FIGS. **6** and **7**, in the backlight unit **5** in the related art, a positioning hole **49** is also provided in the first support portion **43** of the lens **40**. Two positioning pins **69** are provided in the lower case **60**, and both of the two positioning pins **69** are respectively inserted into the positioning hole **49** of the first support portion **43** and the positioning hole **47** of the second support portion **45**. In such a configuration, the lens **40** is held by the case **70**.

(34) In the example in the related art, since the positioning hole **49** is also provided in the first support portion **43**, the first support portion **43** is increased in size, which leads to an increase in size of the lens **40** and the backlight unit **5**. In contrast, in the vehicular display device **1** according to the embodiment, the lens **40** is reduced in size as described above, and thus the backlight unit **5** can be reduced in size.

(35) The present disclosure is not limited to the embodiments described above, and various modifications can be adopted within the scope of the present disclosure. For example, the present disclosure is not limited to the embodiments described above, and modifications, improvements, and the like can be appropriately made. In addition, materials, shapes, sizes, numbers, arrangement positions and the like of components in the embodiments described above are freely selected and are not limited as long as the present disclosure can be implemented.

(36) Although the first support portion **43** to be fitted into the support portion fitting space **R** is

provided on one side of the lens body **41**, and the second support portion **45** having the positioning hole **47** is provided on the other side of the lens body **41** in the embodiment described above, the present disclosure is not limited thereto. The lens **40** may have a support portion that does not have a positioning hole and is to be fitted into a space provided in the case **70** on one side as well as on the other side of the lens body **41**.

(37) Although an example in which the case **70** is completed by combining the upper case **50** and the lower case **60** and the lens **40** is accommodated in the case **70** is described in the embodiment described above, the present disclosure is not limited thereto. The lens may be accommodated inside a case in which an upper case and a lower case are integrated. In this case, for example, a space capable of accommodating a first support portion of the lens is provided in a part of the integrated case, and an opening through which the lens can be inserted is provided in a surface opposite to a surface in which the space is provided. According to such a configuration, the lens can be held in the case by inserting the lens into the case through the opening and inserting the first support portion into the space.

(38) Here, features of embodiments of the display unit and the vehicular display device according to the present disclosure described above are briefly summarized and listed in the following aspects 1 to 5.

(39) According to an aspect 1 of the present disclosure, a display unit (**5**) emits a display image including display light to be projected onto a projection target member (**103**) provided in a vehicle (**100**). The display unit includes: a lens (**40**) including a lens body (**41**) and a first support portion (**43**) protruding from an edge portion of the lens body; and a case (**70**) that has a space (R) configured to accommodate the first support portion and holds the lens by fitting the first support portion into the space.

(40) According to the display unit having the configuration of the aspect 1, the lens is positioned and held in the case by fitting the first support portion into the space of the case. Accordingly, as compared with a case where positioning holes are provided in support portions of the lens and the lens is assembled in a manner of fitting positioning pins provided in the case into the positioning holes, an area of a support portion can be reduced by a size of a positioning hole. Therefore, it is not necessary to ensure this area, the lens can be reduced in size, and thus the display unit can be reduced in size.

(41) Since no positioning hole is provided in the support portion (the first support portion) of the lens, formability of the lens is improved, and the degree of freedom in designing a shape of the lens is increased. This makes it easy to adopt an optimum shape for improving visibility. Further, it is possible to achieve a Fresnel lens which cannot be achieved in a lens in the related art in which positioning holes are provided in support portions. Therefore, the lens can be further reduced in size and thickness by adopting the Fresnel lens.

(42) When a plurality of lenses are used to improve visibility, the display unit may be increased in size, whereas according to the above configuration, since the lens can be reduced in size, a plurality of lenses can be used without increasing the size of the display unit, and display quality of the vehicular display device can be improved.

(43) According to an aspect 2 of the present disclosure, in the aspect 1, the case includes a first case (**50**) and a second case (**60**), and the space is defined by combining the first case and the second case.

(44) According to the display unit having the configuration of the aspect 2, the first support portion of the lens is positioned between the first case and the second case, the case sandwiches the first support portion in the space by combining the first case and the second case, so that the lens can be easily assembled to the case.

(45) According to an aspect 3 of the present disclosure, in the aspect 2, the first case has a first side wall (**51**) having an opening (**53**). The second case has a second side wall (**62**) provided with a placement surface (**64**) on which the first support portion is placed. The placement surface is

provided with a pair of position regulating portions (66) configured to regulate a position of the first support portion on the placement surface. The space is defined by the opening, the placement surface, and the pair of position regulating portions in a state where the first case and the second case are combined.

(46) According to the display unit having the configuration of the aspect 3, since a position of the first support portion placed on the placement surface is regulated by the pair of position regulating portions, assembling accuracy of the lens to the case can be improved.

(47) According to an aspect 4 of the present disclosure, in the aspect 2, the lens body deflects light incident from a light source and has a rectangular shape when viewed in an optical axis direction. The first support portion is provided on one side of the lens body. A second support portion (45) having a positioning hole (47) is provided on the other side of the lens body opposite to the one side. A positioning portion (69) to be inserted into the positioning hole is provided in either the first case or the second case.

(48) According to the display unit having the configuration of the aspect 4, the first case and the second case are combined in a state where the first support portion on one side of the rectangular lens body is fitted into the space and the positioning portion is inserted into the positioning hole in the second support portion on the other side. Therefore, the lens can be reduced in size as compared with a case where positioning holes are provided in support portions on one side and the other side of the lens body.

(49) According to an aspect 5 of the present disclosure, a vehicular display device (1) includes: the display unit according to any one of the aspects 1 to 4; and an optical component (8, 9) configured to reflect the display light emitted from the display unit.

(50) According to the vehicular display device having the configuration of the aspect 5, the vehicular display device can be reduced in size and display quality can be improved.

Claims

1. A display unit that emits a display image including display light to be projected onto a projection target member provided in a vehicle, the display unit comprising: a lens including a lens body and a first support portion protruding from an edge portion of the lens body; a case that has a space configured to accommodate the first support portion and holds the lens by fitting the first support portion into the space, the case including: a first case including a first side wall having an opening; and a second case including a second side wall provided with a placement surface on which the first support portion is placed, the placement surface being provided with a pair of position regulating portions configured to regulate a position of the first support portion on the placement surface, the first support portion being disposed in the space between the pair of position regulating portions, wherein the space is defined by the opening, the placement surface, and the pair of position regulating portions in a state where the first case and the second case are combined.

2. A vehicular display device comprising: the display unit according to claim 1; and an optical component configured to reflect the display light emitted from the display unit.

3. A display unit that emits a display image including display light to be projected onto a projection target member provided in a vehicle, the display unit comprising: a lens including a lens body and a first support portion protruding from an edge portion of the lens body; a case that has a space configured to accommodate the first support portion and holds the lens by fitting the first support portion into the space, the case including: a first case including a first side wall having an opening; and a second case including a second side wall provided with a placement surface on which the first support portion is placed, the placement surface being provided with a pair of position regulating portions configured to regulate a position of the first support portion on the placement surface, the first support portion being disposed in the space between the pair of position regulating portions, wherein the space is defined by combining the first case and the second case, the lens

body deflects light incident from a light source and has a rectangular shape when viewed in an optical axis direction, the first support portion is provided on one side of the lens body, and a second support portion having a positioning hole is provided on the other side of the lens body opposite to the one side.

4. The display unit according to claim 3, wherein a positioning portion to be inserted into the positioning hole is provided in either the first case or the second case.

5. A vehicular display device comprising: the display unit according to claim 3; and an optical component configured to reflect the display light emitted from the display unit.
